Michele Bilko

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EDUCATION

Tufts University, Medford, MA Sep 2026 – May 2028
M.S. in Computer Science · Expected Graduation: 05/28
Relevant Coursework: Data Structures & Algorithms, Advanced Algorithms, Software Engineering, Full-Stack Development, Database Systems, Cloud Computing, Networks

Boston University, Boston, MA Sep 2024 – May 2026
B.A. in Computer Science, Minor in Philosophy · Dean's List · Graduated 05/26

University of California, Santa Cruz, CA Sep 2023 – Jun 2024
B.S. coursework in Computer Science & Game Development (transferred) · Dean's Honors List

INTERNSHIP EXPERIENCE

Intern — Columbia University Media Center for Art History Sep 2022 – Jun 2023

- Engineered an automated data-processing pipeline (Java, Python, pandas) transforming 20,000+ lines of imaging metadata into a normalized, searchable schema — the finding aid is still in active use by the Media Center today.
- Applied photogrammetry and Structure-from-Motion 3D-reconstruction algorithms to build virtual tours from archival image sources, and contributed to the design and ongoing maintenance of the Media Center's public-facing website.

RESEARCH EXPERIENCE

Bike-Routing Infrastructure Research — Tufts University (Incoming) Fall 2026 – Present
Advised by Prof. Justin Hollander

- Joining a project to build a deep-learning-based bike-routing and navigation model, applying spatial and urban-planning data to model and recommend rideable infrastructure.

Directed Study in Graph Algorithms — Boston University Jan 2026 – Present
Advised by Prof. Dora Erdős

- Investigated graph conductance and network connectivity; built a Python/NetworkX framework benchmarking distributed conductance-testing algorithms (BTT vs. Fichtenberger–Vasudev) across varied topologies. Finalizing manuscript for submission.

Satellite Image Restoration Research — Boston University Jan 2026 – May 2026

- Built a deep-learning pipeline (PyTorch) reconstructing ~22% of stratified data gaps in Landsat 7 imagery, extending the LaMa inpainting model with multi-temporal reference guidance and CBAM attention.

TECHNICAL PROJECTS

Massachusetts Rivers Alliance · Tufts JumboCode · React, Node.js, PostgreSQL Sep 2026 – Present

- Joined a student-run JumboCode development team building a full-stack digital tool for the Massachusetts Rivers Alliance to support watershed and river-health tracking, volunteer coordination, and policy-advocacy work across the state's rivers.

Central Rock Gym Route Tracking System · React, TypeScript, Django, PostgreSQL Aug 2025 – Dec 2025

- Designed, built, and deployed a full-stack web application end-to-end — React/TypeScript frontend, Django backend with custom ORM queries and triggers, real-time data synchronization, and an interactive D3.js analytics dashboard.

Boston Music Zine Website · Freelance Web Development 2026 – Present

- Designing and building a website for a local Boston house-show music zine end-to-end, managing scope and client feedback directly.

ADDITIONAL EXPERIENCE

Student Technician (BU IT Help Center) · Front Desk Staff (Central Rock Gym Fenway) · Bike Shop Technician (REI) Aug 2024 – Present

- Balanced technical support, customer service, and daily operations across three concurrent part-time and seasonal roles — diagnosing issues, translating technical concepts for non-technical users, and keeping fast-paced environments running smoothly.

LEADERSHIP & ACTIVITIES

- Founded Generation Coders, a high-school coding club teaching Java and Python to students at schools without CS offerings.
- Member: JumboCode, Tufts Trading Club, Graduate Women in Engineering, Tufts Graduate Student Council.
- Authored the Wikipedia article on Fine-Grained Complexity, synthesizing technical results for a broad audience; wrote a graduate-level survey paper on the same topic (CS 530).

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, C#, HTML/CSS

Frameworks & Libraries: React, Django, Node.js, Express, Flask, PyTorch, TensorFlow, scikit-learn, NumPy, pandas, SQLAlchemy, NetworkX, D3.js

Systems & Tools: Git/GitHub, Docker, Linux/Unix, PostgreSQL, RESTful API design, Agile, unit & integration testing, LaTeX

AI Tools: Fluent with LLM-based coding assistants and agents (GitHub Copilot, Cursor, Claude) for prototyping and workflow acceleration.